

Whitepaper Document

LTIMindtree Specialized Knowledge Transfer for Capital Group CDM ETL Modernization

Employee	Nikhil Subhash Shinde
Employer	LTIMindtree
Current Position	Senior Specialist - Data Engineering
Proposed U.S. Role	Senior Data Engineer / Technical Lead - Canvas AIOps, Generative AI and Cloud Data Modernization Specialist
Primary Project	Capital Group - CDM ETL Modernization
Business Unit / Horizontal Unit	BFS / Data & Analytics
Purpose	Attorney-review whitepaper supporting L-1B specialized-knowledge positioning

1. Executive Summary

Primary L-1B Whitepaper Theme

Nikhil Shinde should be positioned as an LTIM specialized-knowledge resource, not as a generic Python/Spark engineer. His value is the integrated transfer of LTIM proprietary Canvas knowledge, LTIM modernization delivery methods, Capital Group CDM implementation context, and self-created Python migration accelerators used to reduce conversion and validation risk.

This whitepaper summarizes why the proposed U.S. assignment requires specialized knowledge of LTIMindtree products, delivery methods, processes, tools, and client-specific modernization context. The role supports continuity of Capital Group CDM ETL modernization by preserving legacy Informatica transformation logic, applying repeatable Python/PySpark migration patterns, and accelerating source-to-target validation.

Legal framing: For L-1B review, this document is written around company-specific or advanced knowledge rather than general industry skill. Final legal positioning should be aligned by LTIM immigration counsel with the petition letter, manager support letter, organization chart, and evidence exhibits.

2. Business and Technical Problem Statement

Business Problem	Why Specialized Knowledge Is Required
Legacy Informatica dependency	The CDM platform contains legacy Informatica workflows, sessions, mappings, parameters, transformation logic, and dependency chains that must be understood before modernization.
Risk of business-rule loss	Migration is not a simple code rewrite. It requires preserving source-to-target intent, transformation rules, delta logic, reconciliation controls, and financial-data quality expectations.
Need for reusable delivery	LTIM must deliver repeatable modernization patterns rather than one-off scripts, including framework components for ingestion, validation, logging, auditability, and deployment readiness.
Need for LTIM platform alignment	The assignment requires use of LTIM Canvas AIOps and Canvas Insights concepts for operations intelligence, impact analysis, traceability, testing impact, and delivery acceleration.

3. Organization and Reporting Context

The current organization chart places Nikhil Shinde in the BFS business unit and Data & Analytics horizontal unit. The chart shows Nikhil as Senior Specialist - Data Engineering within the architecture-led data engineering delivery structure, reporting through the Principal Architecture chain and coordinating with delivery data engineers.

Current Organization Chart Detail	Value
Business Unit	BFS
Horizontal Unit	Data & Analytics
Duration	From July 2024 - Till date
Team Distribution	US Based: 6 India Based: 12 Total shown: 20
Nikhil Role Shown	Nikhil Shinde - Senior Specialist, Data Engineering
Direct Reporting Chain	Venu Lambu -> Krishnan Iyer -> Sriram N -> Aniruddha Liladhar Vaidya -> Arnab Basu -> Arindam Dutta -> Rahul Banerjee -> Nikhil Shinde

Current Organization Chart Snapshot

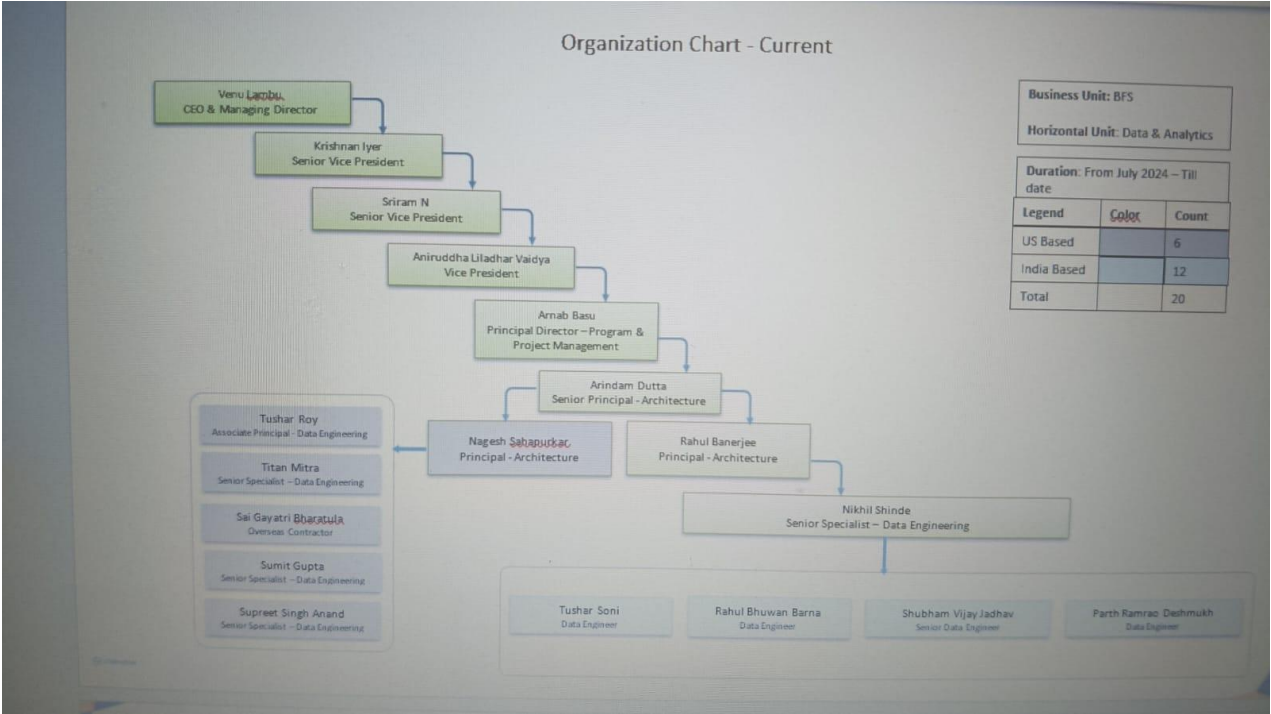


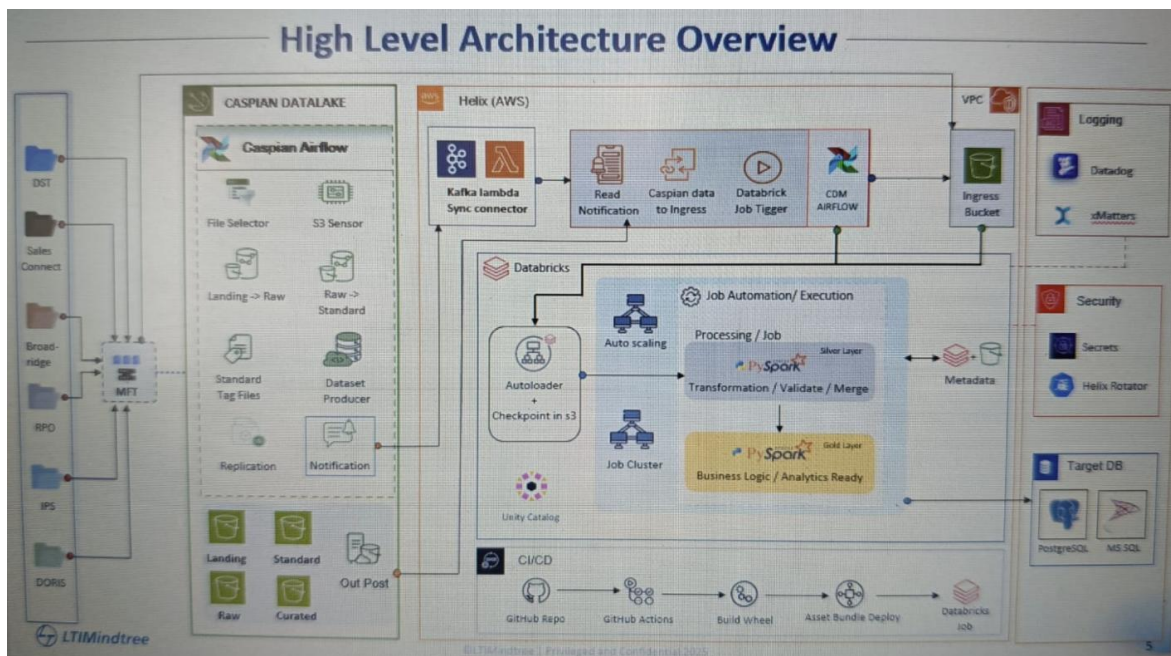
Figure 1. Current organization chart supplied for whitepaper alignment. Use original/source version in final petition exhibits.

4. Capital Group CDM Modernization Solution Overview

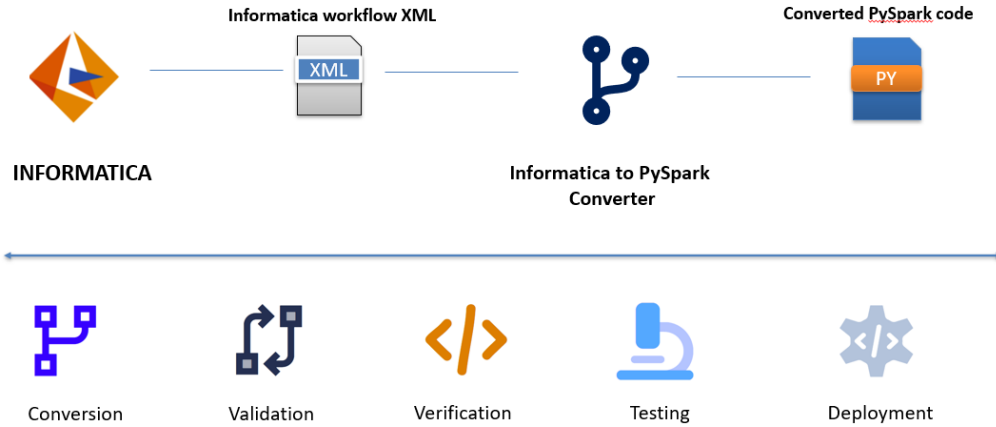
Modernization Layer	Whitepaper Description
Assessment	Analyze Informatica PowerCenter XML assets, workflows, sessions, mappings, sources, targets, parameters, transformation rules, and dependencies.
Conversion	Convert legacy Informatica logic into modular Python/PySpark patterns while preserving business rules and transformation intent.
Framework	Create reusable components for ingestion, transformation, delta handling, validation, reconciliation, publishing, configuration, JDBC connectivity, logging, audit tracking, and exception handling.
Orchestration	Support Airflow-based orchestration and Databricks/AWS/EKS-aligned execution patterns for modernized pipelines.
Validation	Use FastRecon and source-to-target controls to compare legacy outputs with modernized pipeline outputs across files, SQL tables, and SQL queries.
Operations	Align production-readiness with Canvas AIOps concepts such as alert correlation, CI/service impact, SOP recommendation, monitoring, and proactive anomaly handling.

5. Specialized Knowledge Required

Specialized Knowledge Area	Applied Role Requirement
Capital Group CDM context	Understand CDM modernization objectives, workflow dependencies, transformation preservation, validation expectations, and client-specific deployment context.
LTIM modernization methods	Apply LTIM-style legacy assessment, Informatica conversion patterns, reusable framework design, orchestration migration, quality controls, and release readiness.
Canvas AIOps	Use LTIM AIOps concepts for AI-led operations, alert convergence, causal analysis, noise suppression, ITSM integration, CI impact, Jovi GenAI assistance, digital SOPs, and anomaly detection.
Canvas Insights	Use LTIM SDLC intelligence concepts for knowledge fabric, requirement-code-defect correlation, impact analysis, test optimization, defect advisor, solution advisor, and reuse insights.
Python accelerators	Apply FastRecon, informatica-python, and informatica-sparker for reconciliation, Informatica XML parsing, metadata extraction, and PySpark conversion acceleration.
Financial data engineering	Work with fixed income/equity data, reconciliation controls, governance, reporting, data quality, and enterprise investment-data processing expectations.



Accelerating Informatica to PySpark Migration



70%

Automated Coverage

Auto-converts most Informatica mapping logic into PySpark draft code.

4-5 Hour

Time Saved/Mapping

Cuts manual rewrite time with faster analysis, code generation, and templating.

30%

Assisted Finalization

Reviewer-led tuning for edge cases, performance, and code readiness.

Faster ETL Modernization

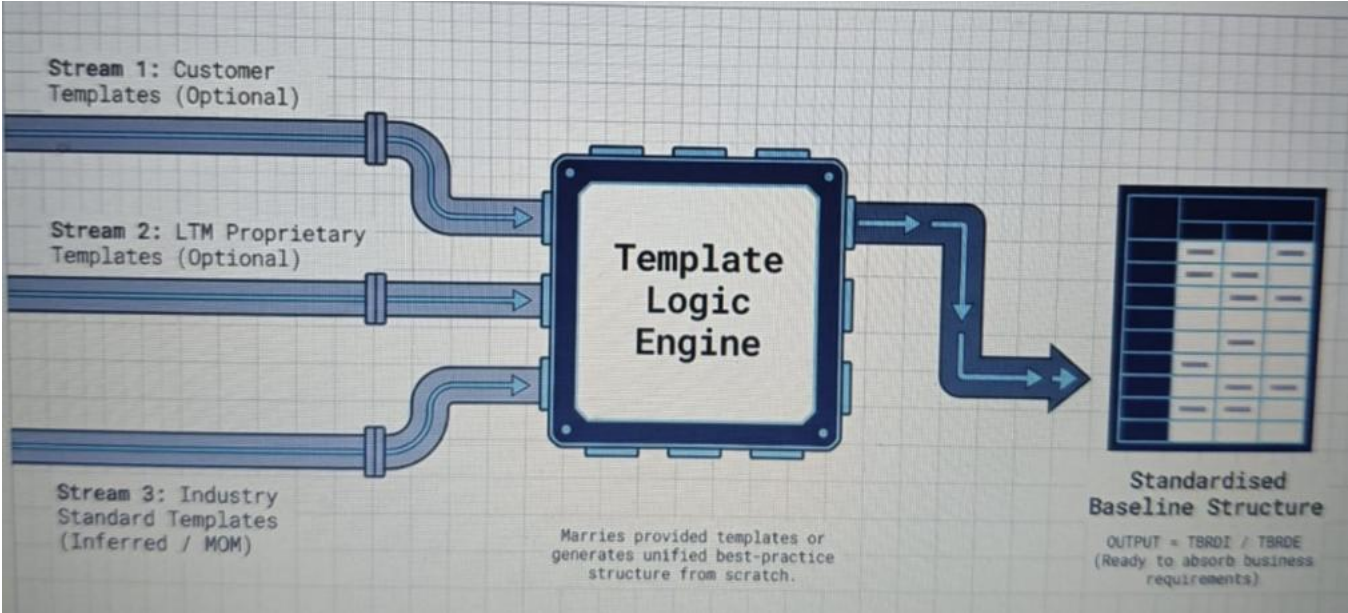
Automated **Informatica to PySpark code drafts** conversion with expert-guided finalization for accurate, consistent migrations.

The screenshot shows the fastrecon website interface. The top navigation bar includes links for Intro, Install, Matrix, Sources, Modes, Config, Examples, API, FAQ, and About. A search bar is on the right. The left sidebar lists 33 conversion types, such as '20. JSON → SQL Query' and '33. Every compare_mode Mode'. The main content area features a 'What Python version does fastrecon require?' section, an 'About fastrecon' section, and a profile for Nikhil Shinde. The bottom of the page includes tags for various data tools and licenses, and links to external resources.

6. Key Role Duties with Percentage Allocation

Duty Area	%	Whitepaper Role Description
CDM modernization architecture and conversion strategy	25%	Lead design for Informatica-to-Python/PySpark modernization; interpret workflows, mappings, sessions, dependencies, source-target logic, and transformation rules.
Reusable Python/PySpark modernization framework	20%	Design standardized components for ingestion, transformation, delta processing, validation, reconciliation, publishing, configuration, logging, audit tracking, and error handling.
Metadata extraction and conversion	15%	Apply informatica-python and informatica-sparker to extract Informatica metadata and

accelerators		accelerate Spark/PySpark-aligned conversion patterns.
Validation and reconciliation controls	15%	Use FastRecon and source-to-target validation patterns to compare legacy and modernized outputs across files, SQL tables, and SQL queries.
Canvas-enabled SDLC and operations alignment	15%	Apply Canvas Insights for traceability and impact analysis; apply Canvas AIOps for production-readiness, incident intelligence, digital SOPs, and proactive monitoring.
Leadership, governance, and delivery coordination	10%	Guide data engineers, review designs/code, align with architecture/program management, QA, governance, and stakeholders, and support release readiness.
Total	100%	Integrated role requiring LTIM-specific, CDM-specific, and accelerator-specific knowledge.



7. Self-Created Tools and Specialized Value

Tool / Library	Purpose	L-1B Specialized Value
FastRecon	Python reconciliation library for comparing files, SQL tables, SQL queries, and heterogeneous migration outputs.	Reduces manual validation effort and strengthens source-to-target proof for legacy-to-cloud migration.
informatica-python	Python library for parsing Informatica PowerCenter XML and extracting workflow/session/mapping metadata.	Transforms legacy ETL knowledge into repeatable modernization analysis capability.
informatica-sparker	Python/PySpark-oriented accelerator for generating Spark-aligned structures from Informatica metadata.	Accelerates Informatica-to-PySpark conversion and standardizes migration output.

8. Why This Cannot Be Readily Replaced by a General Market Hire

General Market Capability	Remaining Gap Without Nikhil
Python, Spark, or cloud development experience	Does not provide immediate knowledge of Capital Group CDM workflow intent, Informatica dependencies, LTIM conversion patterns, and reusable framework design.
Generic ETL migration experience	Does not provide Nikhil's hands-on CDM modernization context, source-to-target validation approach, or migration accelerators.
Observability or SDLC tool experience	Does not provide LTIM Canvas AIOps / Canvas Insights platform knowledge or how LTIM positions them for modernization and production support.
Ability to build scripts	Does not provide already-authored FastRecon, informatica-python, and informatica-sparker assets or the project knowledge embedded in them.
Financial data engineering background	May not include Capital Group CDM-specific modernization logic, workflow dependency semantics, or LTIM governance and release practices.

9. Business Benefit to LTIM U.S. Assignment

- Accelerates U.S. delivery ramp-up by transferring existing LTIM/CDM modernization knowledge rather than rebuilding it locally.
- Reduces migration risk by preserving legacy transformation intent and applying proven source-to-target validation controls.
- Improves repeatability by reusing Python/PySpark framework patterns and self-created migration accelerators.
- Strengthens client confidence through continuity across architecture, conversion, validation, orchestration, and production-readiness workstreams.
- Connects LTIM proprietary Canvas AIOps and Canvas Insights capabilities to real CDM modernization and operations use cases.

10. Evidence and Exhibit Checklist

Priority	Evidence	Purpose
P1	Manager support letter	Confirms specialized LTIM/CDM knowledge and why the U.S. role requires Nikhil.
P1	Current/proposed organization chart	Shows role placement within BFS / Data & Analytics reporting structure.
P1	Capital Group CDM artifacts	Architecture diagrams, conversion design, sanitized Informatica workflow samples, framework notes, validation plan.
P1	Canvas AIOps / Canvas Insights evidence	Training/course screenshots, internal notes, workshop material, demo records, or manager confirmation.
P1	FastRecon / informatica library evidence	PyPI/GitHub links, screenshots, documentation, commit history, usage examples.
P2	Performance appraisal / promotion records	Supports technical leadership, architecture ownership, AI skills, and project impact.
P2	Resume and HR employment proof	Supports role history and qualifying employment background.

11. Whitepaper Conclusion

Final Positioning Statement

Nikhil Shinde's proposed assignment should be framed as a specialized LTIM/CDM modernization transfer. The role requires LTIM proprietary Canvas knowledge, Capital Group CDM implementation context, Informatica-to-Python/PySpark modernization methods, and reusable Python accelerators. This combination is materially different from general data engineering ability and supports the business need for his U.S. assignment.

Source/legal note: This draft is prepared for attorney review and is not legal advice. Legal framing should be finalized by LTIM immigration counsel. General L-1B specialized-knowledge language should be aligned with USCIS Policy Manual Volume 2, Part L, Chapter 4, 8 C.F.R. 214.2(l), and 8 U.S.C. 1184(c)(2).